SAMPLE

Superconductivity

Zero Resistance and the Future of Energy

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Introduction

Superconductivity is a set of physical properties observed in superconductors: materials where electrical resistance is exactly zero and magnetic fields are expelled from the material. Unlike an ordinary metallic conductor, whose resistance decreases gradually as its temperature is lowered, even down to near absolute zero, a superconductor has a characteristic critical temperature below which the resistance drops abruptly to zero. An electric current through a loop of superconducting wire can persist indefinitely with no power source.

The superconductivity phenomenon was discovered in 1911 by Dutch physicist Heike Kamerlingh Onnes. Like ferromagnetism and atomic spectral lines, superconductivity is a phenomenon which can only be explained by quantum mechanics. It is characterized by the Meissner effect, the complete cancellation of the magnetic field in the interior of the superconductor during its transitions into the superconducting state. The occurrence of the Meissner effect indicates that superconductivity cannot be understood simply as the idealization of perfect conductivity in classical physics.

In 1986, it was discovered that some cuprate-perovskite ceramic materials have a critical temperature above 35 K (−238 °C). It was found shortly thereafter (by Ching-Wu Chu) that replacing the lanthanum in the recipe with yttrium, making YBCO, pushed the critical temperature above 90 K. Such a high transition temperature is theoretically impossible for a conventional superconductor, leading the materials to be termed high-temperature superconductors.

The cheaply available coolant liquid nitrogen boils at 77 K (−196 °C) and thus the existence of superconductivity at higher temperatures than this facilitates many experiments and applications that are less practical at lower temperatures. Superconductivity was discovered in solid mercury on April 8, 1911, by Heike Kamerlingh Onnes, who was studying the resistance of mercury at cryogenic temperatures using the recently produced liquid helium as a refrigerant. At the temperature of 4.2 K, he observed that the resistance abruptly disappeared.

In 1935, Fritz and Heinz London showed that the Meissner effect was a consequence of the minimization of the electromagnetic free energy carried by superconducting current.

The Londons wrote their insight as two constitutive equations for a superconductor: the first follows from Newton's second law for the superconducting electrons, and the second asserts that any magnetic field caught inside decays exponentially with depth, over the short range now called the London penetration depth.

Together the two equations showed that superconductivity is a quantum state with a genuine rigidity against fields, not merely a state of zero resistance.

However, it is clear that a two-electron pairing is involved, although the nature of the pairing -- whether s-wave, d-wave, or something more exotic -- is a question the field still argues about.

Chapter 1: The Discovery of Superconductivity

Great efforts have been devoted to finding out how and why superconductivity works; a decisive clue came in 1933, when Meissner and Ochsenfeld discovered that superconductors expelled applied magnetic fields, the phenomenon now known as the Meissner effect.

The theoretical model that was first conceived for superconductivity was completely classical: it is summarized by London constitutive equations. It was put forward by the brothers Fritz and Heinz London in 1935, shortly after the discovery that magnetic fields are expelled from superconductors. A major triumph of the equations of this theory is their ability to explain the Meissner effect, wherein a material exponentially expels all internal magnetic fields as it crosses the superconducting threshold.

By using the London equation, one can obtain the dependence of the magnetic field inside a superconductor on the distance to the surface: the field dies away exponentially, so that in a sample much thicker than the penetration depth the interior never feels an applied field at all.

During the 1950s, theoretical condensed matter physicists arrived at an understanding of "conventional" superconductivity through a pair of theories: the phenomenological Ginzburg-Landau theory (1950) and the microscopic BCS theory (1957). In 1950, the phenomenological Ginzburg-Landau theory of superconductivity was devised by Landau and Ginzburg. This theory, which combined Landau's theory of second-order phase transitions with a Schrödinger-like wave equation, had great success in explaining the macroscopic properties of superconductors.

In particular, Abrikosov showed that Ginzburg-Landau theory predicts the division of superconductors into the two categories now referred to as Type I and Type II. Abrikosov and Ginzburg were awarded the 2003 Nobel Prize for their work (Landau had received the 1962 Nobel Prize for other work, and died in 1968). The four-dimensional extension of the Ginzburg-Landau theory, the Coleman-Weinberg model, is important in quantum field theory and cosmology. Also in 1950, Maxwell and Reynolds et al.

This important discovery pointed to the electron-phonon interaction as the microscopic mechanism responsible for superconductivity. The complete microscopic theory of superconductivity was finally proposed in 1957 by Bardeen, Cooper and Schrieffer. This BCS theory explained the superconducting current as a superfluid of Cooper pairs, pairs of electrons interacting through the exchange of phonons. For this work, the authors were awarded the Nobel Prize in 1972.

The BCS theory was set on a firmer footing in 1958, when N. N. Bogolyubov showed that the BCS wavefunction, which had originally been derived from a variational argument, could be obtained using a canonical transformation of the electronic Hamiltonian. In 1959, Lev Gor'kov showed that the BCS theory reduced to the Ginzburg-Landau theory close to the critical temperature.

Generalizations of BCS theory for conventional superconductors form the basis for the understanding of the phenomenon of superfluidity, because they fall into the lambda transition universality class. The extent to which such generalizations can be applied to unconventional superconductors is still controversial. The first practical application of superconductivity was developed in 1954 with Dudley Allen Buck's invention of the cryotron. Two superconductors with greatly different values of the critical magnetic field are combined to produce a fast, simple switch for computer elements.

Soon after discovering superconductivity in 1911, Kamerlingh Onnes attempted to make an electromagnet with superconducting windings but found that relatively low magnetic fields destroyed superconductivity in the materials he investigated. Much later, in 1955, G. B. Yntema succeeded in constructing a small 0.7-tesla iron-core electromagnet with superconducting niobium wire windings. Then, in 1961, J. E. Kunzler, E. Buehler, F. S. L. Hsu, and J. H. Wernick broke the impasse.

Wernick made the startling discovery that, at 4.2 kelvin, niobium-tin, a compound consisting of three parts niobium and one part tin, was capable of supporting a current density of more than 100,000 amperes per square centimeter in a magnetic field of 8.8 tesla. The alloy was brittle and difficult to fabricate, but niobium-tin proved useful for generating magnetic fields as high as 20 tesla.

In 1962, T. G. Berlincourt and R. R. Hake discovered that more ductile alloys of niobium and titanium are suitable for applications up to 10 tesla. Commercial production of niobium-titanium supermagnet wire immediately commenced at Westinghouse Electric Corporation and at Wah Chang Corporation. Although niobium-titanium boasts less-impressive superconducting properties than those of niobium-tin, niobium-titanium became the most widely used "workhorse" supermagnet material, in large measure a consequence of its very high ductility and ease of fabrication.

However, both niobium-tin and niobium-titanium found wide application in MRI medical imagers, bending and focusing magnets for enormous high-energy-particle accelerators, and other applications. Conectus, a European superconductivity consortium, estimated that in 2014, global economic activity for which superconductivity was indispensable amounted to about five billion euros, with MRI systems accounting for about 80% of that total. In 1962, Josephson made the important theoretical prediction that a supercurrent can flow between two pieces of superconductor separated by a thin layer of insulator.

In the same experiment, he also observed the superfluid transition of helium at 2.2 K, without recognizing its significance. The precise date and circumstances of the discovery were only reconstructed a century later, when Onnes's notebook was found. In subsequent decades, superconductivity was observed in several other materials. In 1913, Tin and lead were found to superconduct at 3.8 K and 7 K respectively, and in 1941 niobium nitride was found to superconduct at 16 K.